

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1<sup>st</sup> Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination Dec 2016

PHPT001, Novel Drug Delivery System-I

Date: 16.12.2016 , Friday

Time: 10:00 am to 1:00 pm

Maximum Marks: 70

## Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

### SECTION-I

- Q.1 (A) Give the salient features of fast dissolving tablets. 05
- (B) Give the drug selection criteria for fast dissolving tablets with drug examples. 05
- Q.2 (A) Explain the anatomy and physiology of mucus membrane. 05
- (B) Enumerate mucoadhesion theories. Explain any one in brief. 05

### OR

- Q.2 Write a note on intra uterine implants. 10
- Q.3 Answer the following (Any three) 15
- (A) Give the advantages and disadvantages of oral controlled drug delivery system over conventional drug delivery system.
- (B) Controlled drug delivery system can improve the therapeutic efficacy and safety of a drug. – justify it.
- (C) Explain the merits and demerits of the effervescent floating drug delivery system.
- (D) Give the broad classification of pharmaceutical approaches of gastro retentive drug delivery system.

### SECTION-II

- Q.4 (A) Write a Short Note: Stimuli sensitive Hydrogels 05
- (B) Applications of Hydrogels in drug delivery. 05
- Q.5 Give Classification of permeation enhancers and enlist the methods to increase the permeation through skin and describe any one. 10

### OR

- Q.5 Write in detail about Self Emulsifying Drug Delivery system (SMEEDS). 10

- (A) Describe the characterization of nanoparticles.
- (B) Enlist the methods for preparation of liposomes, explain any one in detail.
- (C) Application of solid lipid nanoparticles.
- (D) Give category and example of Excipients used for preparing liposomes, solid lipid nanoparticles and nanostructured lipid carriers.